much earlier, more clement times. The geological odometer was reset. All accessible records of surface geology begin with the last global magma flood. Before they cool and solidify, oceans of lava may be hundreds or even thousands of kilometers thick. In our time, billions of years later, the surface of such a world may be quiet, inactive, with no hint of current vulcanism. Or there may be—as on Earth—a few small-scale but active reminders of an epoch when the entire surface was flooded with liquid rock.

In the early days of planetary geology, ground-based telescopic observations were all the data we had. A fervent debate had been running for half a century on whether the craters of the Moon were due to impacts or volcanos. A few low mounds with summit calderas were found—almost certainly lunar volcanos. But the big craters—bowl- or pan-shaped and sitting on the flat ground and not the tops of mountains—were a different story. Some geologists saw in them similarities with certain highly eroded volcanos on Earth. Others did not. The best counterargument was that we know there are asteroids and comets that fly past the Moon; they must hit it sometimes; and the collisions must make craters. Over the history of the Moon a large number of such craters should have been punched out. So if the craters we see are not due to impacts, where then are the impact craters? We now know from direct laboratory examination of lunar craters that they are almost entirely of impact origin. But 4 billion years ago this little world, nearly dead today, was bubbling and churning away, driven by primeval vulcanism from sources of internal heat now long gone.

In November 1971, NASA's *Mariner 9* spacecraft arrived at Mars to find the planet completely obscured by a global dust storm. Almost the only features to be seen were four circular spots rising out of the reddish murk. But there was something peculiar about them: They had holes in their tops. As the storm cleared, we were able to see unmistakably that we had been viewing four huge volcanic mountains penetrating through the dust cloud, each capped by a great summit caldera.

After the storm dissipated, the true scale of these volcanos became clear. The largest—appropriately named Olympus Mons, or Mt.